

METU Computer Engineering

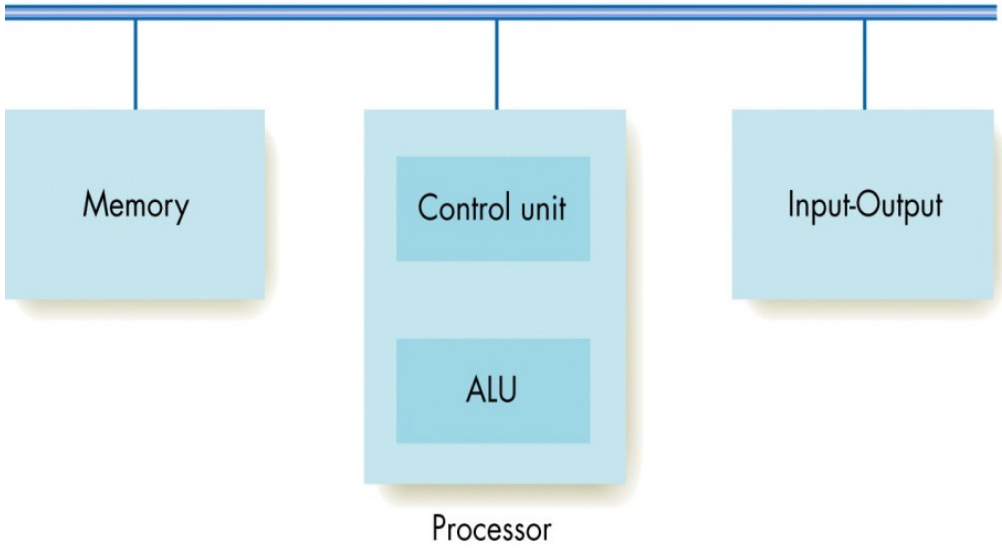
CEng 111 – Fall 2026
Week 3b

Credit: Some slides are from the “Invitation to Computer Science” book by G. M. Schneider, J. L. Gersting and some from the “Digital Design” book by M. M. Mano and M. D. Ciletti.

Previously on CENG111!

Central Processing Unit (CPU)

- Control Unit
- Arithmetic Logic Unit (ALU)



Previously on CENG111!

The Control Unit



- Manages stored program execution
- Task

1. Fetch from memory the next instruction to be executed
2. Decode it: determine what is to be done
3. Execute it: issue appropriate command to ALU, memory, and I/O controllers

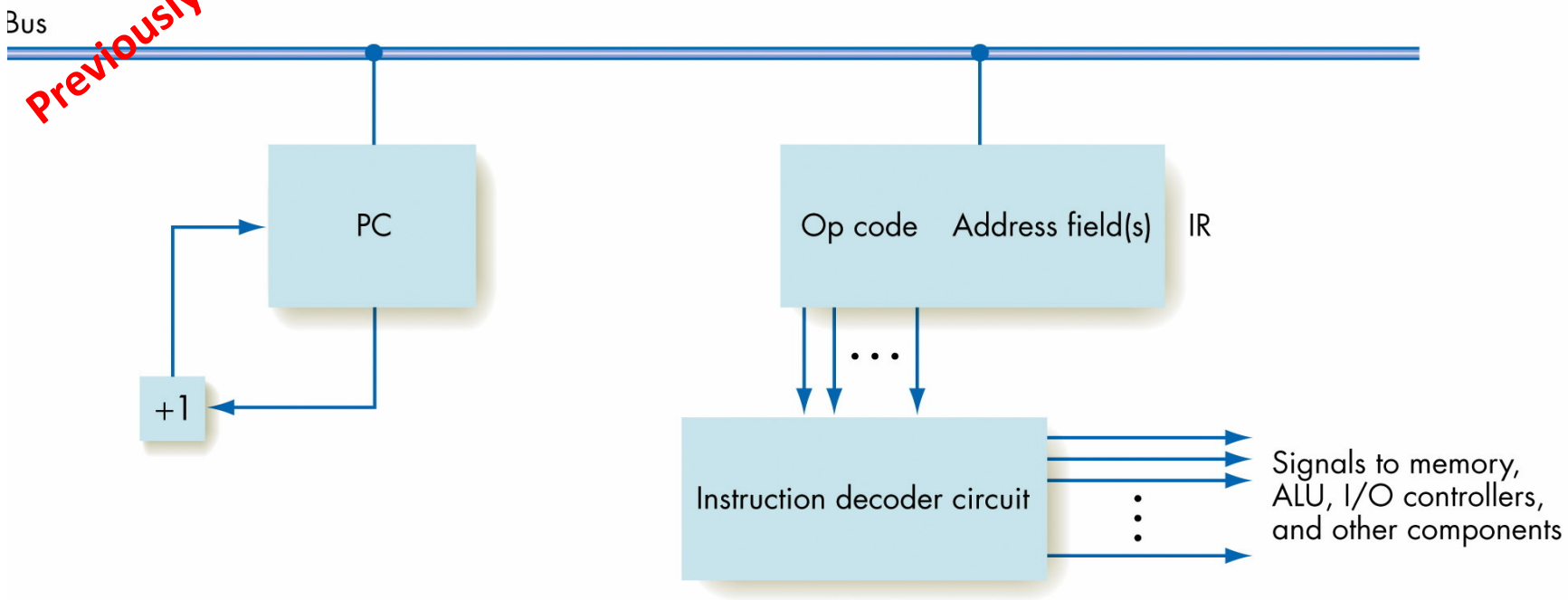
Previously on CENG111!

More on Instructions



- LOAD X -> Load register R with the contents of memory cell X
- STORE X -> Store register R into memory cell X
- MOVE X, Y -> Copy the contents of X into Y
- ADD X -> Add contents of X to the contents of R
- ADD X,Y -> Add contents of X to the contents of Y, and put the result in register R
- COMPARE X, Y-> Set GT (greater than), EQ (equal) and LT (less than) condition codes
- JUMP X-> Jump unconditionally to the instruction in cell X
- JUMPGT X-> Jump, if GT=1, to the instruction in cell X
- HALT

Previously on CENG111!



Organization of the Control Unit Registers and Circuits



Today

- Continue with the Von Neumann Architecture
  - CPU
    - Control Unit and the Fetch-Decode-Execute Cycle
    - Arithmetic Logic Unit



Administrative Notes

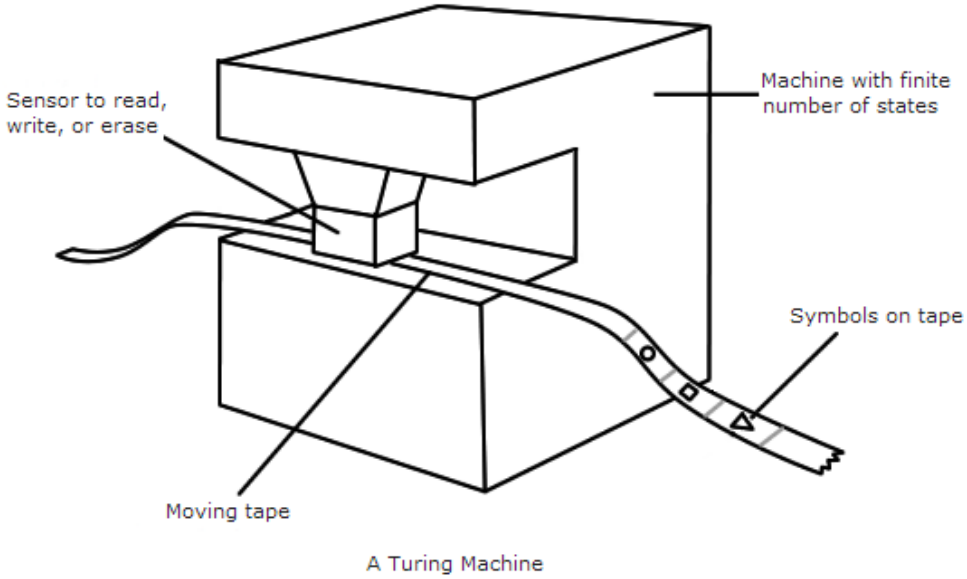
- Labs are starting this week!
- Von Neumann Machine Exam:
  - Preliminary Text has been announced!
  - In-class Exam: 23 October (Friday) – 18:00

Turing Machine



Alan Turing
(1912-1954)

- Main components:
  - Read-write head
  - Infinite memory tape
  - Rule book



A Turing Machine

- It has a set of states and it can recognize a set of symbols

Turing Machine



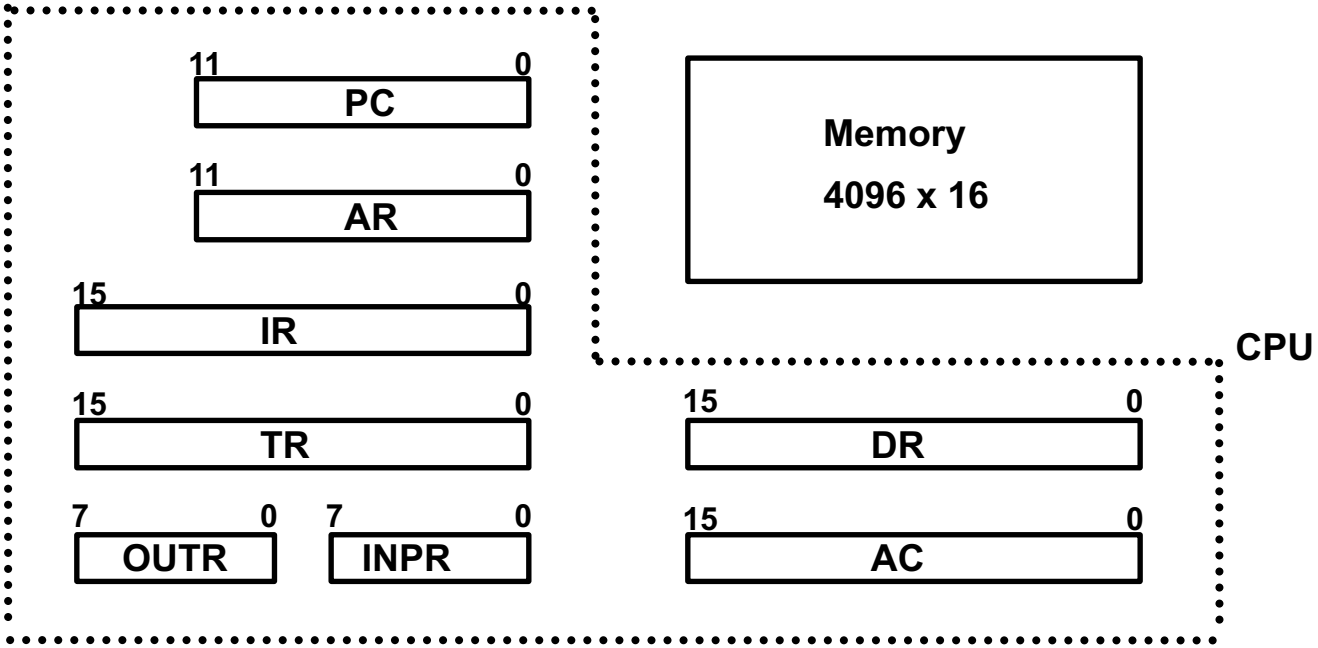
Alan Turing
(1912-1954)

- Follows these steps:
  - Read current symbol on the tape
  - Look at the rule book for that symbol
  - Execute the corresponding rule
- Example (from the second textbook):

State now	Read symbol	Write symbol	Move tape	Next state
A	0	1	RIGHT	**B**
A	1	1	LEFT	**C**
B	0	1	LEFT	**A**
B	1	1	RIGHT	**B**
C	0	1	LEFT	**B**
C	1	1	RIGHT	**HALT**

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Registers in a Basic Computer



List of BC Registers

DR	16	Data Register	Holds memory operand
AR	12	Address Register	Holds address for memory
AC	16	Accumulator	Processor register
IR	16	Instruction Register	Holds instruction
PC	12	Program Counter	Holds address of instruction
TR	16	Temporary Register	Holds temporary data
INPR	8	Input Register	Holds input character
OUTR	8	Output Register	Holds output character

Main Types of Registers

- **User-accessible Registers**:
  - The most common division of user-accessible registers is into data registers and address registers.
- **Data registers**: hold numeric values such as integer and floating-point values.
  - In some older and low end CPUs, a special data register, known as the accumulator, is used implicitly for many operations.
- **Address registers:** hold addresses and are used by instructions that indirectly access memory.
- **Conditional registers:** hold truth values often used to determine whether some instruction should or should not be executed.
- **General purpose registers (GPRs)** can store both data and addresses, i.e., they are combined Data/Address registers.

http://en.wikipedia.org/wiki/Processor_register



Main Types of Registers (cont'd)

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- **Floating point registers** (FPRs) store floating point numbers in many architectures.
- **Constant registers** hold read-only values such as zero, one, or pi.
- **Special purpose registers** (SPR) hold program state; they usually include the program counter (aka instruction pointer), stack pointer, and status register (aka processor status word). In embedded microprocessors, they can also correspond to specialized hardware elements.
- **Instruction registers** store the instruction currently being executed.

http://en.wikipedia.org/wiki/Processor_register



Some Architectures & The number of registers on them

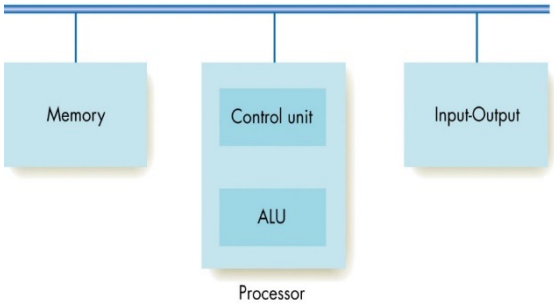
Architecture	Integer registers	FP registers
x86	8	8
x86-64	16	16
IBM/360	16	4
Z/Architecture	16	16
Itanium	128	128
UltraSPARC	32	32
IBM POWER	32	32
Alpha	32	32
6502	3	0
PIC microcontroller	1	0
AVR microcontroller	32	0
ARM	16	16

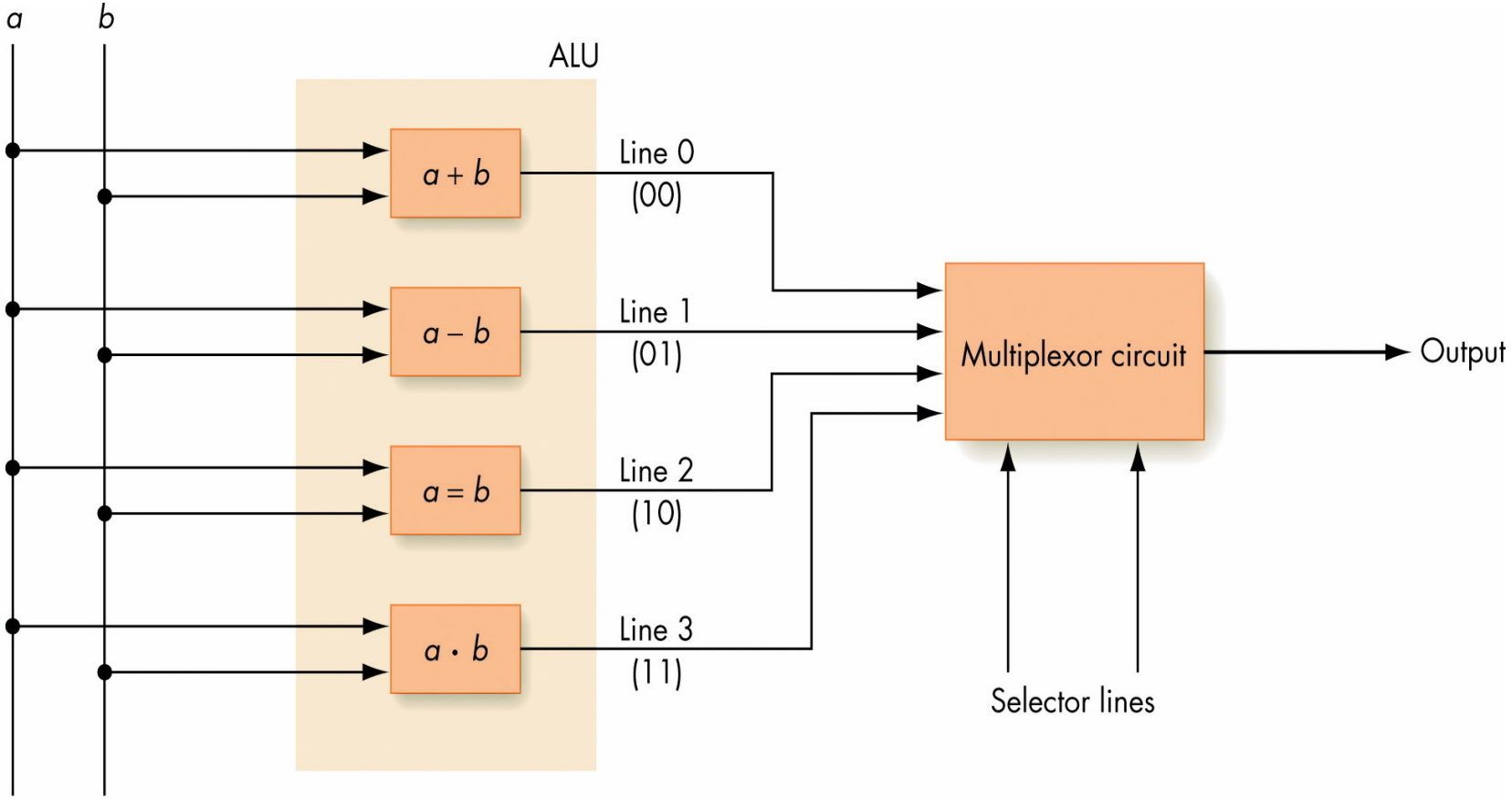
http://en.wikipedia.org/wiki/Processor_register

The Arithmetic/Logic Unit

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- ***A**rithmetic and **L**ogic **U**nit*
  - *"Manufacturing" section of computer*
  - *Contains decision mechanisms and can make calculations+comparisons*
  - Actual computations are performed
- Primitive operation circuits
  - Arithmetic *[+, -, *, /]*
  - Comparison *[equality or CE, GT, LT, NEQ]*
  - Logic *[AND, OR, NOT, XOR]*
- Data inputs and results stored in registers
- Multiplexer selects desired output





Using a Multiplexor Circuit to Select the Proper ALU Result



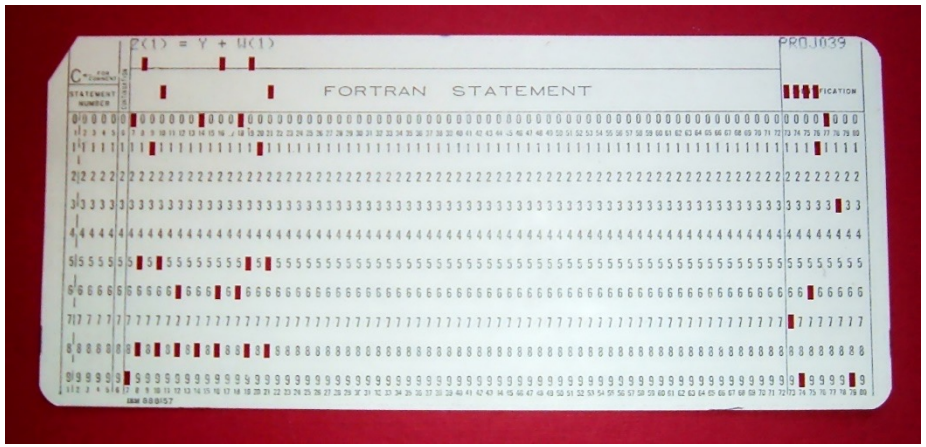
The Arithmetic/Logic Unit (continued)

- ALU process
  - Values for operations copied into ALU's input register locations
  - All circuits compute results for those inputs
  - Multiplexor selects the one desired result from all values
  - Result value copied to desired result register

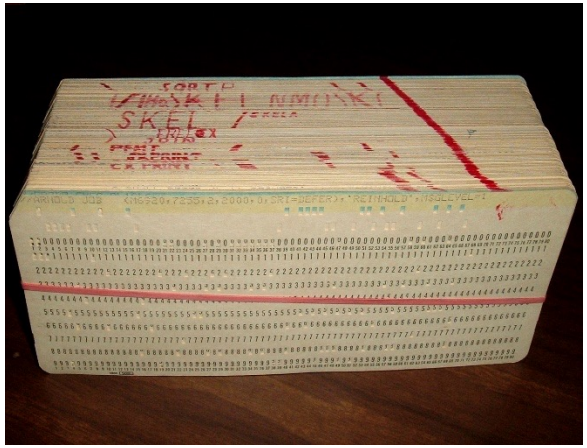


MORE ON
THE VON NEUMANN ARCHITECTURE

The Era of Punched Cards





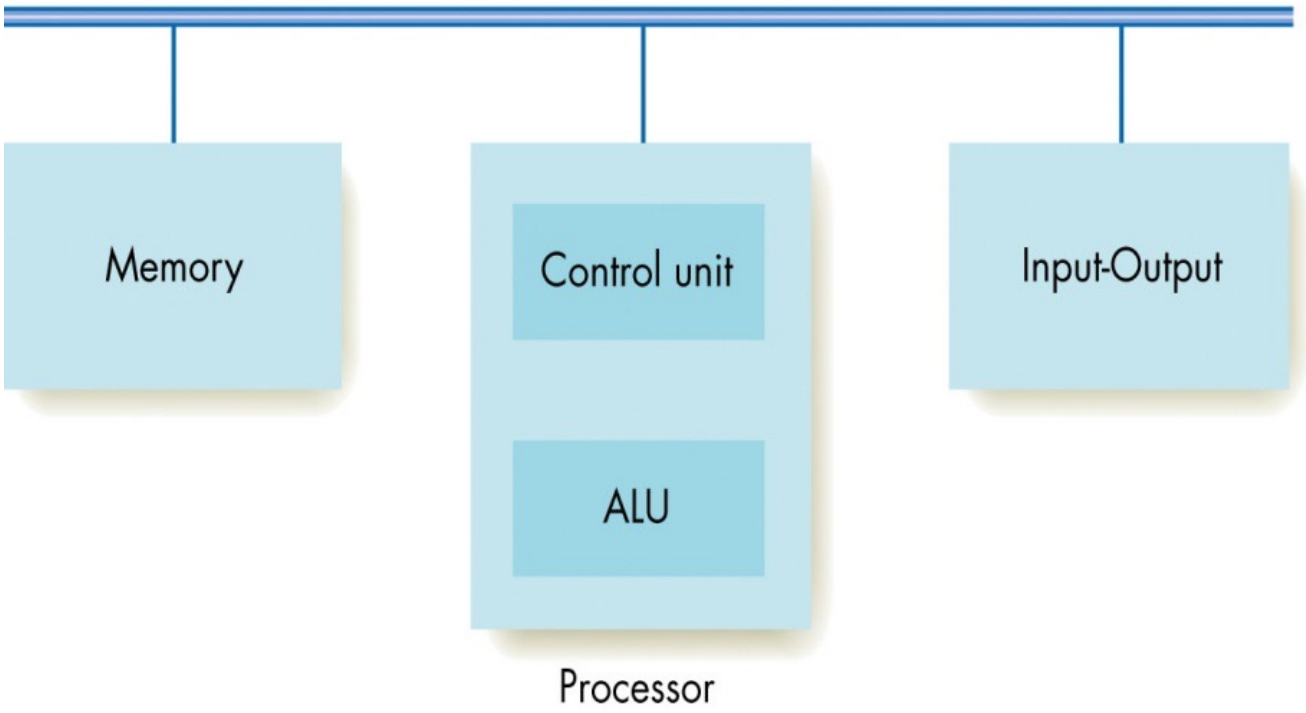


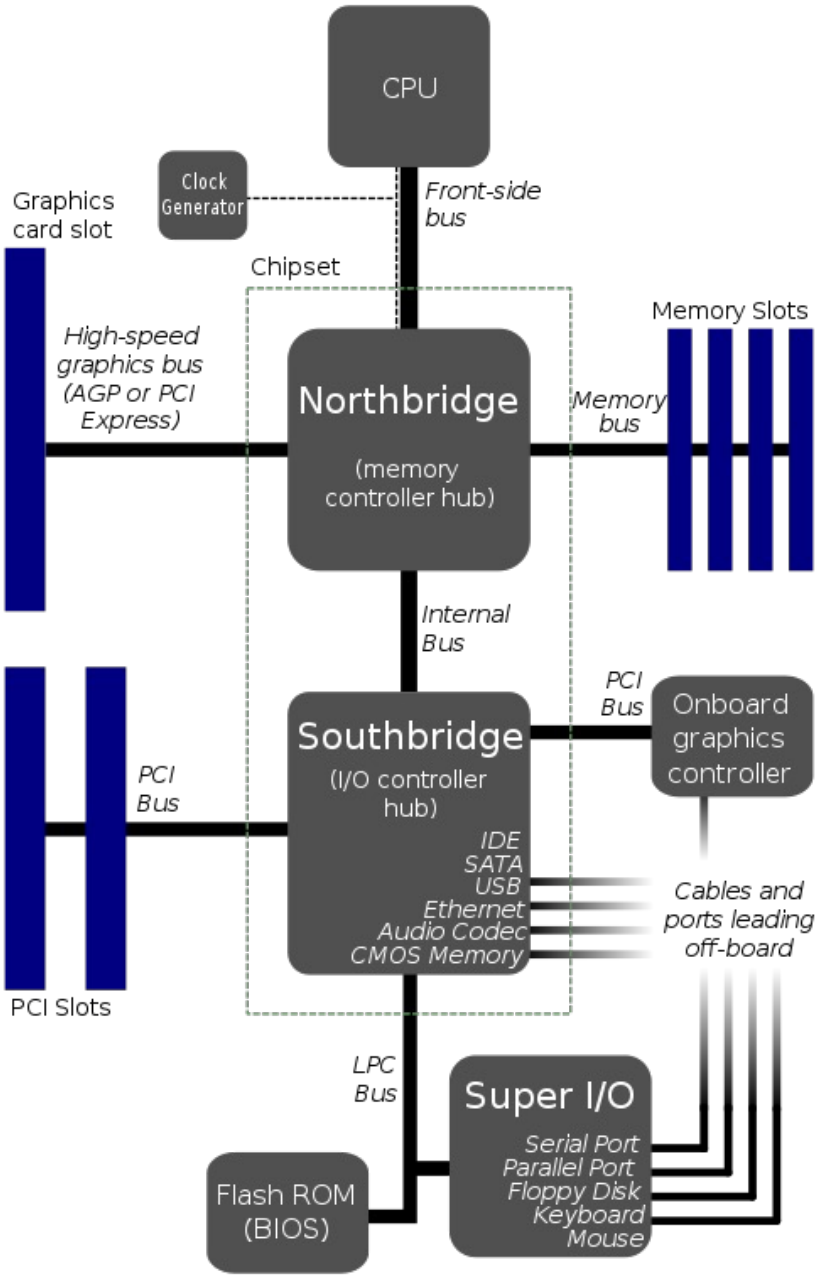
https://en.wikipedia.org/wiki/Computer_programming_in_the_punched_card_era



The Era of Punched Cards

- Machines:
  - Have small memory
  - Some do not have ALU
  - Read programs from the punched card
  - You submit your punch cards, get into a queue and wait for the output
- Example machine:
  - https://en.wikipedia.org/wiki/IBM_1620


Memory
Control unit
ALU
Input-Output
Processor



https://en.m.wikipedia.org/wiki/File:Motherboard_diagram.svg



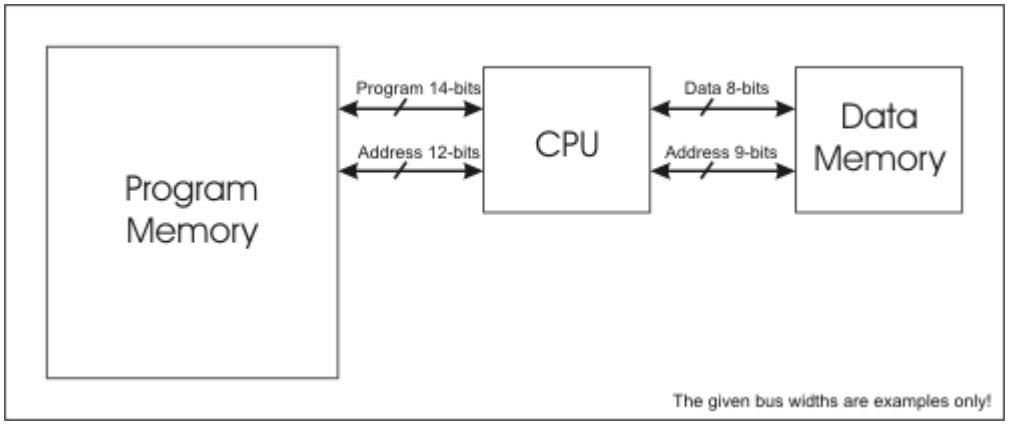
Von Neumann Architecture

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- **Pros:**
  - Simplifies hardware (both circuit-design and layout)
  - Easier to generate re-locatable code, which makes multi-tasking easier to implement.
- **Cons:**
  - Instructions must be multiples of the data bus-width - can be inefficient.
  - Variable number of cycles required for instructions. For example, an instruction that requires data from memory must wait at least another cycle before it can complete, whereas some instructions execute much faster. This can be a problem for time-critical applications.

http://www.mhennessy.f9.co.uk/pic/architecture.htm

Harvard Architecture



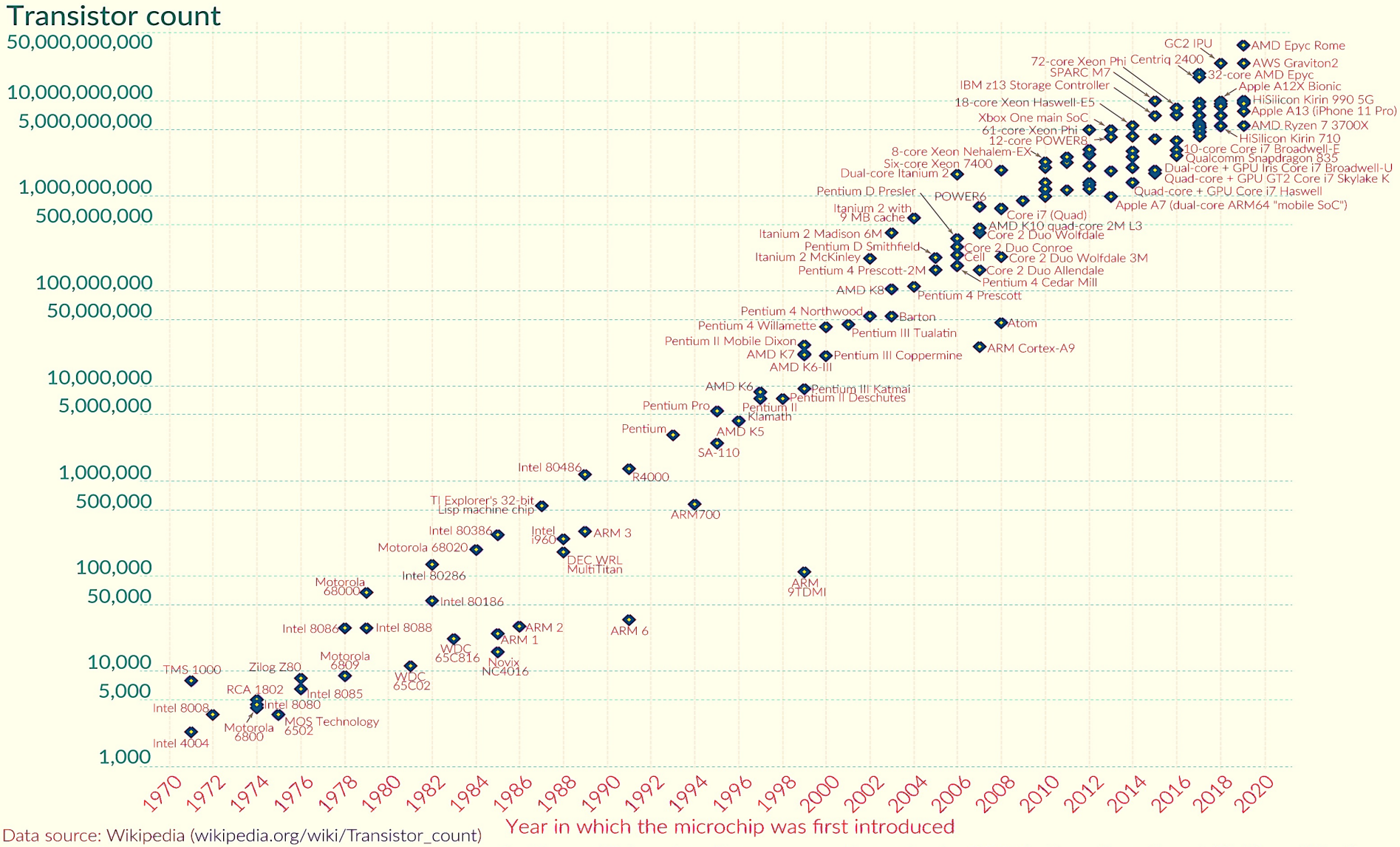
http://www.mhennessy.f9.co.uk/pic/architecture.htm

- **Pros:**
  - Data and address busses can be of different widths. This means the program memory word can be wide enough to incorporate an instruction and a literal (fixed data) in a single instruction.
  - A built-in two-stage pipeline overlaps fetch and execution of instructions, meaning most instructions execute in a single clock cycle.
- **Cons:**
  - Slightly more confusing.
  - Hardware is more complicated.

Moore's Law: The number of transistors on microchips doubles every two years

Our World in Data

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

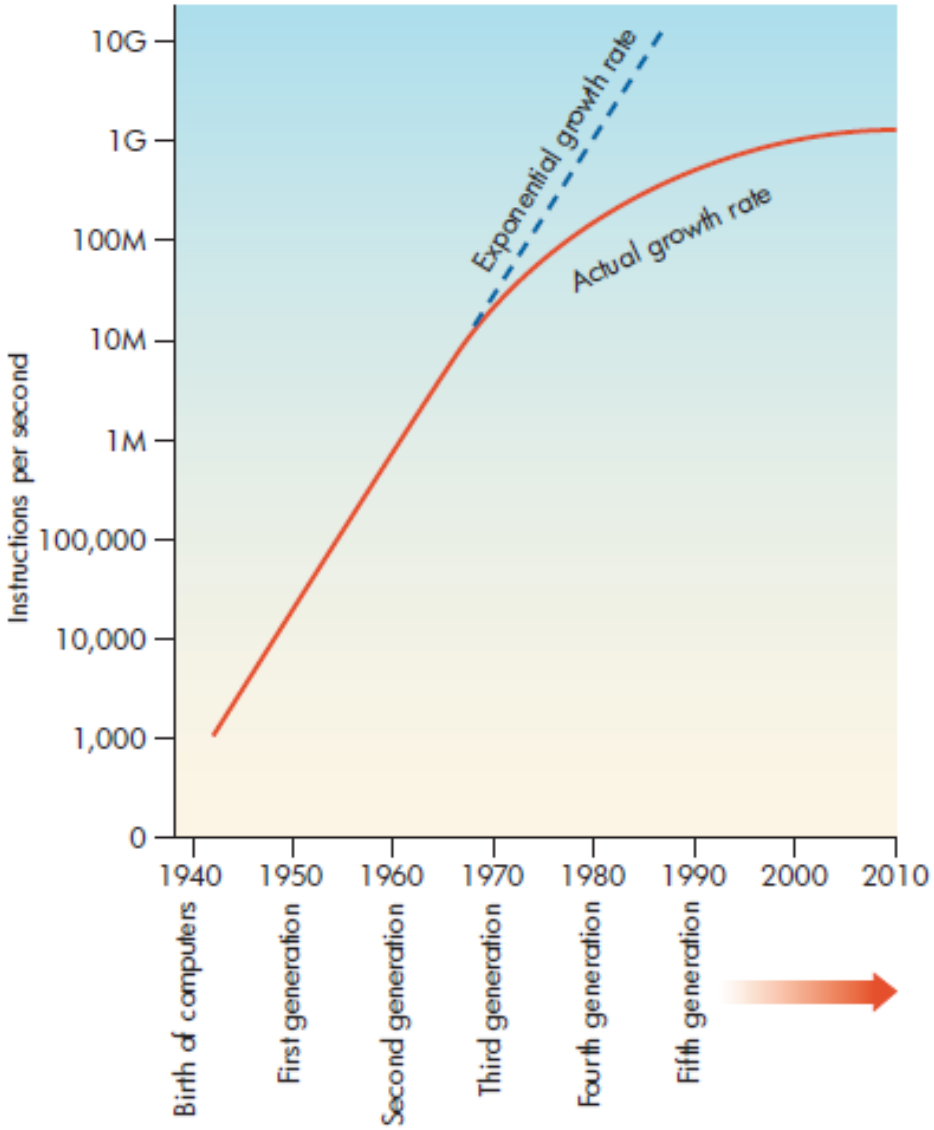


Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)

OurWorldinData.org – Research and data to make progress against the world's largest problems.

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Moore's Law


10G
1G
100M
10M
1M
100,000
10,000
1,000
0
Instructions per second
Exponential growth rate
Actual growth rate
1940
1950
1960
1970
1980
1990
2000
2010
Birth of computers
First generation
Second generation
Third generation
Fourth generation
Fifth generation

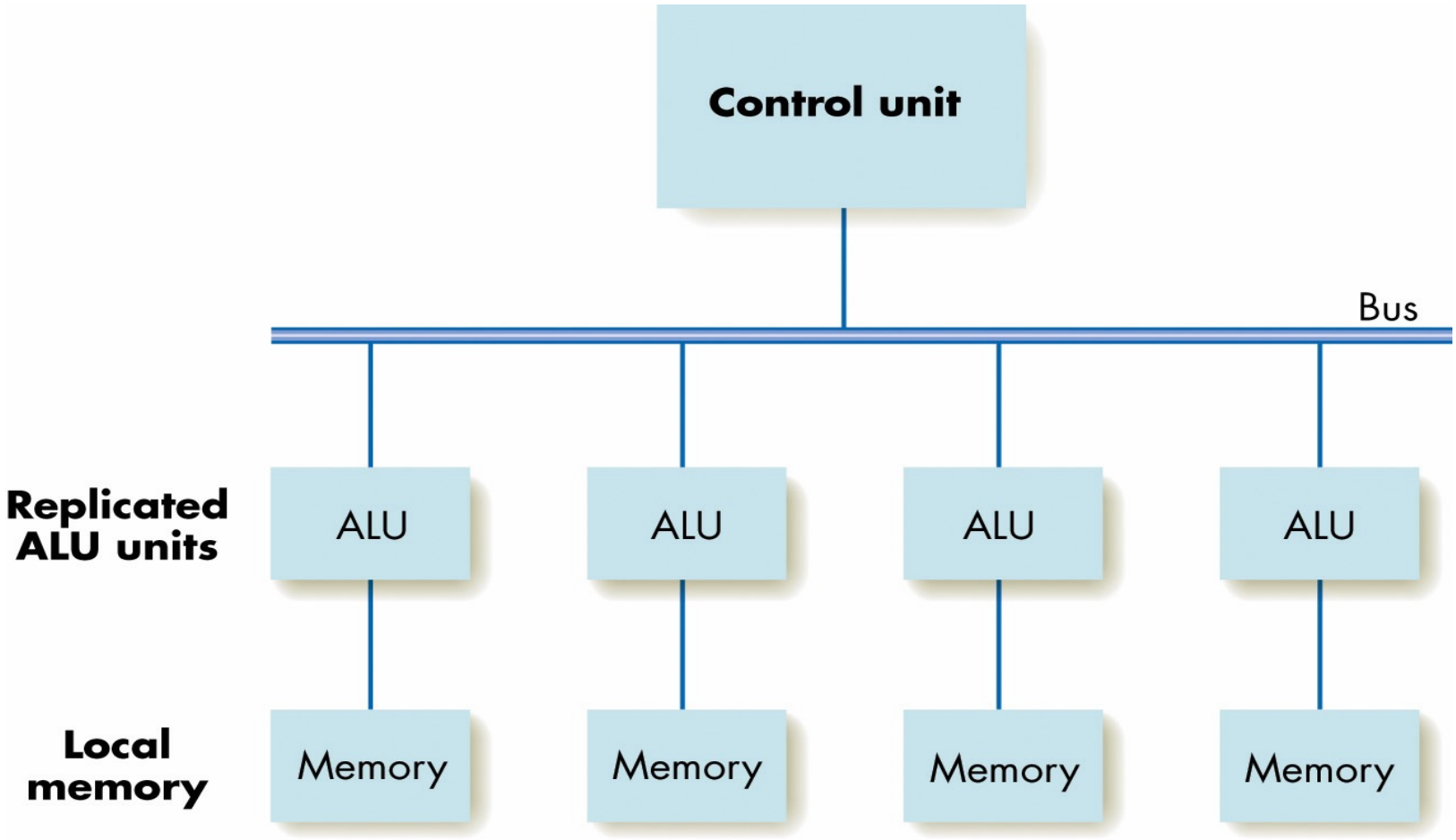
The Future

- Physical limitations on the speed of von Neumann computers
- Alternatives and derivatives of the von Neumann architecture explored to bypass these limitations
- Parallel computing architectures can provide improvements:
  - multiple operations occur at the same time



The Future

- SIMD architecture
  - Single instruction/Multiple data
  - Multiple processors running in parallel
  - All processors execute same operation at one time
  - Each processor operates on its own data
  - Suitable for “vector” operations

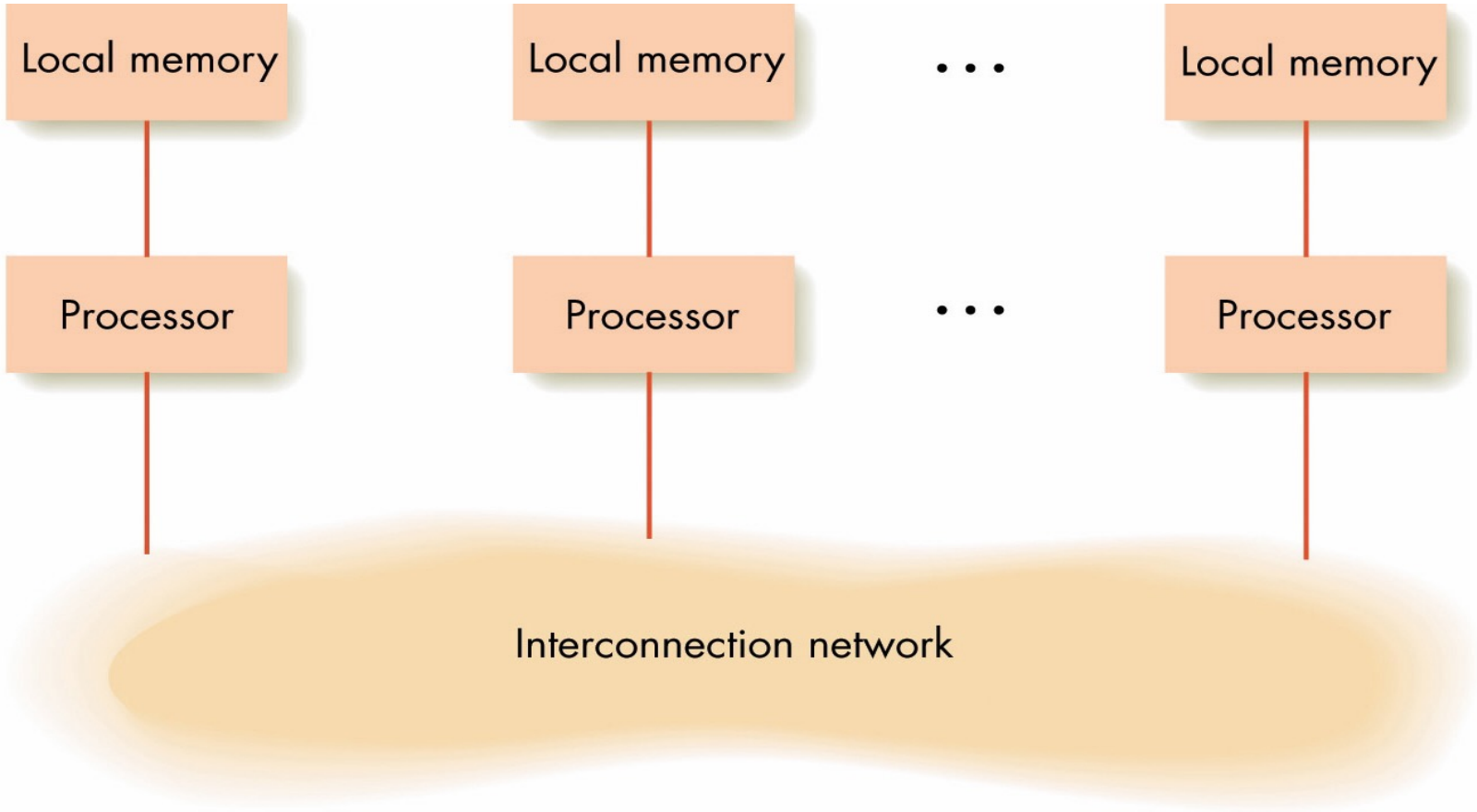


A SIMD Parallel Processing System

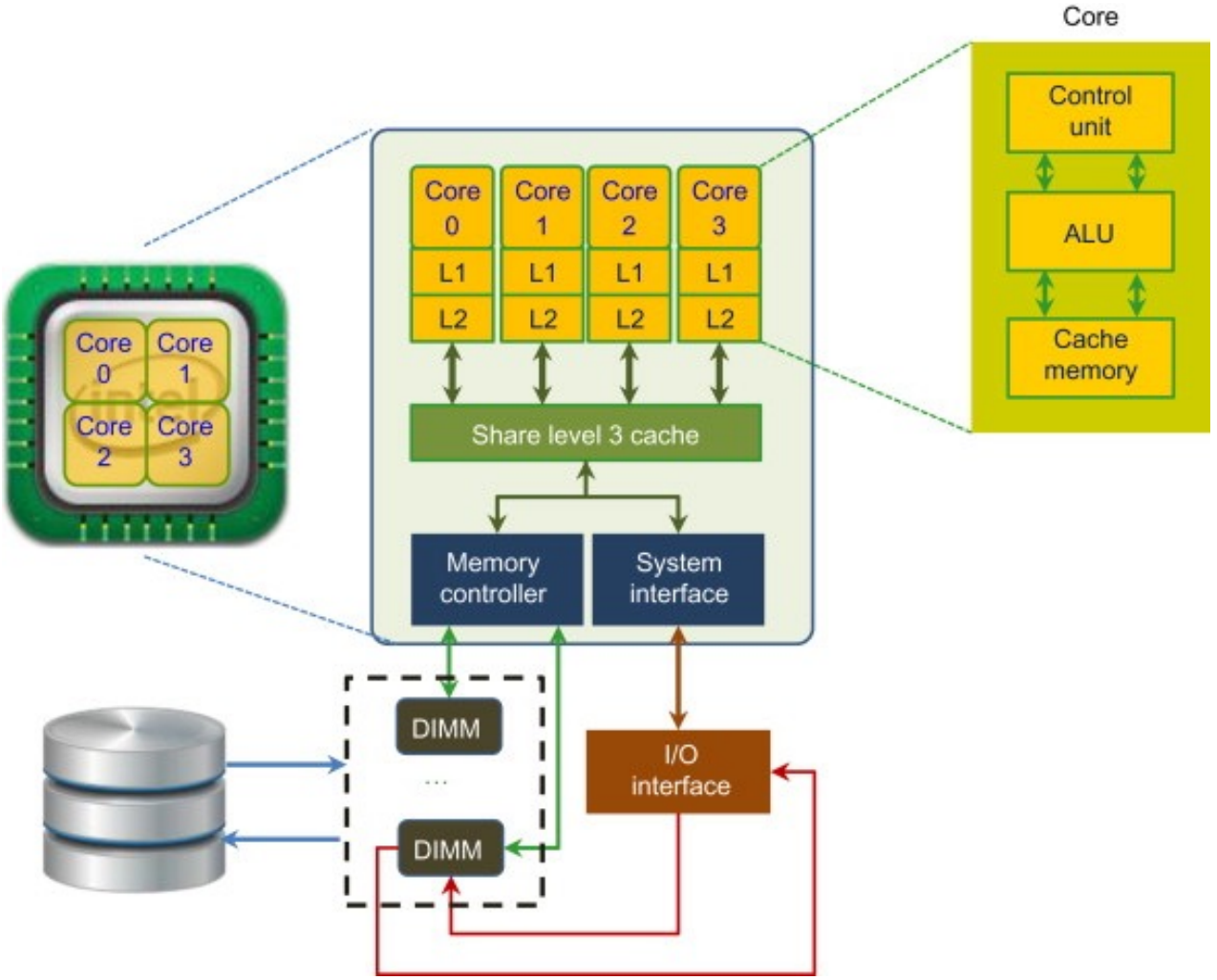


The Future

- MIMD architecture
  - Multiple instruction/Multiple data
  - Multiple processors running in parallel
  - Each processor performs its own operations on its own data
  - Processors communicate with each other



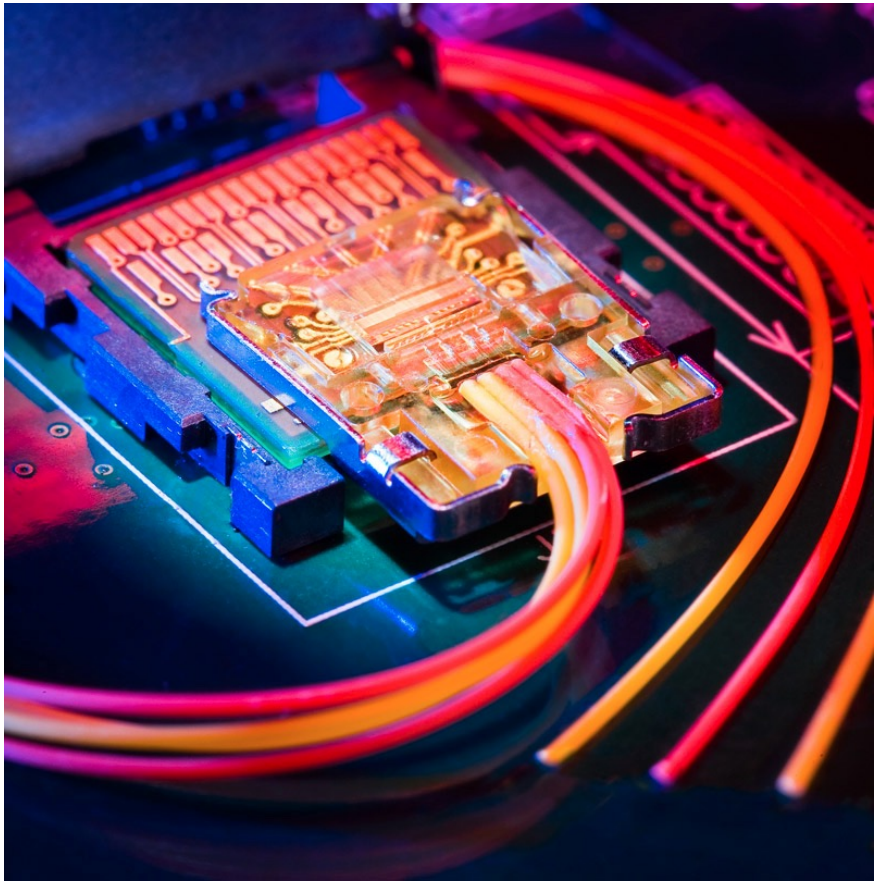
Model of MIMD Parallel Processing



https://www.sciencedirect.com/topics/computer-science/core-processor

New Trend: Optical Computing

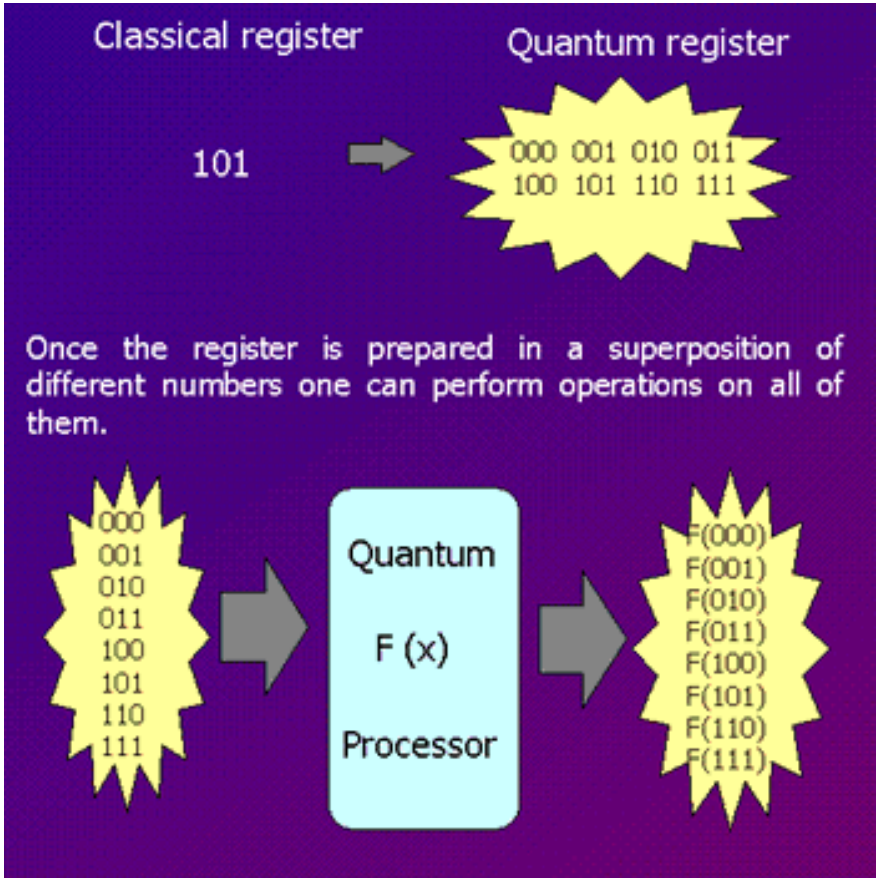
- Currently available for only data transfer.
- Work in progress towards “photonic logic” for designing circuits which use photons:
  - Many challenges



http://en.wikipedia.org/wiki/Optical_computing

New Trend: Quantum Computing





- There are several problems:
  - Decoherence: The more the qubits, the more their effect on the environment.

http://media.defenseindustrydaily.com/images/PUB_CQC_Cambridge_Quantum_Computing_Explained_lg.png

Read Especially This: http://www.cs.virginia.edu/~robins/The_Limits_of_Quantum_Computers.pdf



MORE ON MEMORY



Memory Types

- Dynamic Memory
  - The voltages stored in capacitors die away with time.
  - Solution?
    - Refresh the memory frequently; i.e., re-write the contents of the memory.
- Static Memory
  - A coupled transistor system stores the information.
  - One of the couples *triggers* the other.
- DRAM is cheaper and more widely used.

http://wiki.xtronics.com/index.php/How_Memory_Works



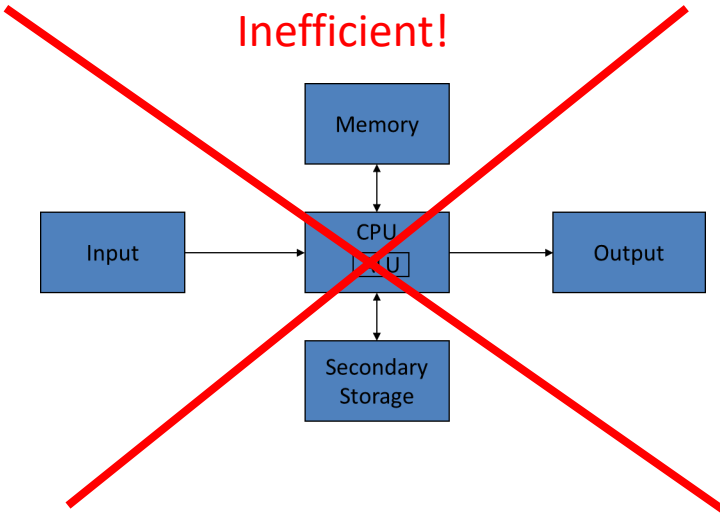
Memory Types

- Volatile Memory:
  - The stored values are lost when power is off.
  - DRAM, SRAM
- Non-volatile:
  - Read-only Memory (ROM), flash memory

http://wiki.xtronics.com/index.php/How_Memory_Works

Direct Memory Access (DMA)

http://camnetway.blogspot.com/2009/11/interrupt-driven-io.html



- All memory accesses go over the CPU.
  - CPU gets the virtual address,
  - It looks up the physical address,
  - It writes the data to the physical device (memory, in most cases)
- This slows down the CPU for simple and consecutive memory accesses.
- "Modern" computers have a channel for DMA.
- The CPU just initiates the DMA command, and a separate unit (DMA controller) handles the copy of data between a device and a memory.

